An experimental Investigation of Oil Churning and Flow Regimes Within a Model Gearbox

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Background

Gear systems have been used widely in the automotive industry due to its potential in achieving efficiencies up to 98~99%. However, for high power systems the power lost could still be in Megawatts(MW). Power loss within a gearbox could be divided into load-dependent(e.g bearing friction) and load-independent(e.g windage, oil churning) power loss. A closer study into the cause of the power loss could allow for better design which reduces power loss and preserve scarce resource.

Aims & Objectives

This study aims to validate existing expressions on oil churning and to investigate the flow regimes within a model gearbox. Several objectives were also outlined:

- Modify existing test rig
- Measure churning torque
- Record flow regimes
- Provide advice on design of new test rig
- Outline areas for future work

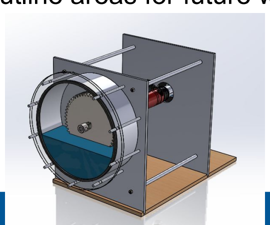


Figure 1: Rendered view of test rig

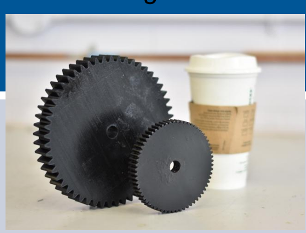


Figure 2: Gears used for testing

Test Rig Modifications

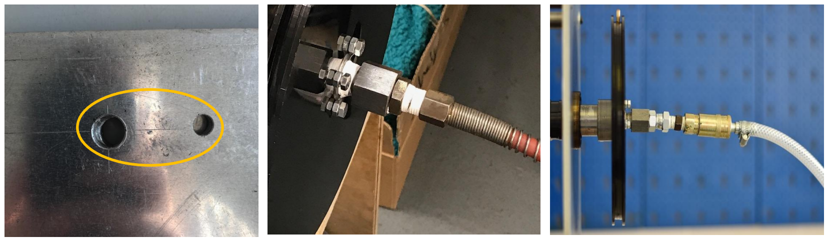
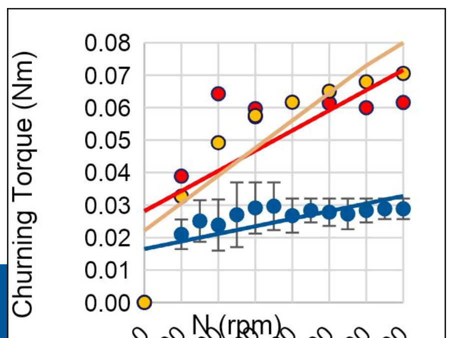
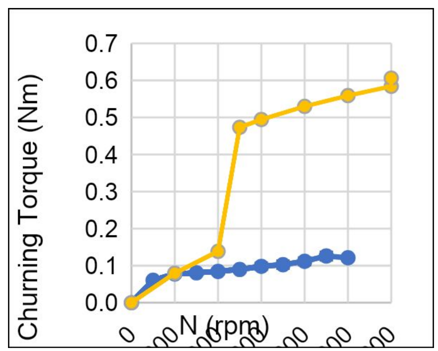


Figure 3: Remanufactured front plate due to alignment issue(left) and addition of a swivel joint to eliminate shear stress on the pulley from twisting the pipe(mid,right).

Oil Churning



Graph 1: Comparison of churning torque measured and expected from Gear 1



Graph 2:Comparsion of churning torque measured and expected from Gear 2

Flow Regimes

Figure 4: Flow regimes captured from Gear 1 and 2 at different immersion ratio and rotational speed

Conclusion

Formulas suggested to express churning torques were rather situational and a more accurate selection on the effective variables are required. The combination of windage and churning has also displayed interesting flow dynamics both above and below the oil level. Findings have also improved the design of the future test rig to include an external reservoir.

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